RealEyes: A Multimodal System for Fake News Detection and AI-Generated Image and Video Forensics with Explainable Decision Reasoning

Rana Moussa[1], Alaaelddin Salama[2], Malak Yasser[3], Eyad Elnakib[4], Mazen Ahmed[5]
Department of Artificial Intelligence Science, Alamein International University, Alamein, Egypt
[1]22101478 [2]22101463 [3]22100400 [4]22100757 [5]22100369

Supervisor: Dr. Mohamed Elkholy
Department of Artificial Intelligence Science, Alamein International University, Alamein, Egypt

***Abstract*—The rapid advancement of generative AI technologies has made it increasingly difficult to distinguish authentic content from synthetically produced or manipulated media. This paper presents RealEyes, a multimodal detection system that addresses two interrelated threats: the spread of textual misinformation and the proliferation of AI-generated or manipulated images, including frame-level analysis of AI-generated video content. The system integrates four specialized components: (1) a Vision Transformer-based classifier for detecting AI-generated images, achieving up to 98.18% validation accuracy using the Swin Transformer V2-Base architecture fine-tuned on the OPENFAKE dataset of 418,524 images; (2) a pixel-level image forgery localization model, AXUNet, which combines an Xception encoder, Transformer self-attention bottleneck, and UNet++ nested decoder trained on 100,000 spliced images; a 3-class natural language inference (NLI) module for automated fact verification (SUPPORTS, REFUTES, NOT ENOUGH INFO), fine-tuning DeBERTa-v3-base on the FEVER and SNLI datasets and achieving 91.86% macro F1; and (4) an OCR pipeline for extracting multimodal content from screenshots. Critically, RealEyes incorporates an explainability layer that translates model decisions into human-readable reasoning, making results accessible to non-technical users. Experimental results demonstrate strong performance across all components, establishing RealEyes as a comprehensive, end-to-end solution for combating AI-generated disinformation.**

***Index Terms*—fake news detection, AI-generated image forensics, image forgery localization, natural language inference, explainable AI, multimodal detection, vision transformers, fact verification, video forensics**

I. Introduction

The emergence of sophisticated generative models, including large-scale diffusion models such as Stable Diffusion and Flux, autoregressive image generators such as GPT Image 1, and large language models, has fundamentally altered the landscape of digital information integrity. Realistic synthetic images can now be produced in seconds from a text prompt, and false factual claims can be amplified at scale across social media platforms before human fact-checkers can respond [11], [22]. The convergence of these threats (visual, textual, and video-based disinformation) demands detection systems that operate across modalities rather than addressing each in isolation.

Existing approaches to this problem are largely unimodal. Prior work on AI-generated image detection, such as [23] and [16], has focused primarily on classifying images as real or fake at the image level, without pinpointing where within an image a manipulation occurred. Meanwhile, fact-verification systems such as those built on BERT [3] and RoBERTa [13] treat misinformation as a purely textual phenomenon, overlooking the visual dimension of modern fake news. Furthermore, most existing detection systems output a simple binary label without providing any explanation that a non-expert user can meaningfully act upon. This lack of transparency erodes user trust and limits real-world adoption.

This paper addresses these limitations through RealEyes, a unified multimodal detection system that combines image classification, pixel-level forgery localization, natural language inference, and explainable AI into a single integrated pipeline. The system is motivated by the observation that fake news in the modern era is rarely purely textual or purely visual; it typically combines fabricated text claims with manipulated or AI-generated supporting imagery. By processing both modalities simultaneously and providing human-readable explanations for each decision, RealEyes represents a step toward practical, deployable disinformation detection.

The primary contributions of this work are as follows:

1) **Multimodal detection architecture.** We propose a five-module system that jointly addresses AI-generated image detection, video-level frame analysis, tampered region localization, and textual claim verification within a single pipeline.
2) **High-accuracy image classification.** We fine-tune three vision transformer architectures (ViT-B/16 in two configurations and Swin Transformer V2-Base) on the large-scale OPENFAKE benchmark, with Swin V2-B achieving 98.18% validation accuracy and an F1 score of 0.9818. An extended higher-capacity variant, Swin V2-B (ms), further achieves 99.02% validation accuracy and an F1 score of 0.9901.
3) **Video-level fake detection.** We extend the image classification module to video content through a frame-sampling pipeline that classifies each frame indepen-

dently and aggregates predictions into a single verdict, enabling detection of fully synthetic and partially manipulated videos.

4) **Pixel-level forgery localization.** We introduce AXUNet, a novel encoder-decoder architecture combining an ImageNet-pretrained Xception encoder, a Transformer self-attention bottleneck, and a UNet++ nested decoder for detecting manipulated regions at the pixel level in spliced images.
5) **Transformer-based fact verification.** We fine-tune BERT, RoBERTa, and DeBERTa-v3-base on a combined FEVER and SNLI corpus of approximately 538,000 examples. DeBERTa-v3-base achieves a macro F1 of 91.86% on the FEVER internal validation set and 91.72% on the blind test set, with strong cross-domain generalization on MNLI.
6) **Explainable output layer.** RealEyes incorporates a human-readable reasoning module that translates model confidence scores and localization maps into plain-language explanations, making detection decisions interpretable for non-technical users.

We begin by situating RealEyes within the existing literature on fake media detection, then describe the datasets, architectures, and training procedures before presenting experimental results and concluding with directions for future work.

II. Related Work

A. AI-Generated Image Detection

The problem of distinguishing real photographs from synthetically generated images has attracted growing attention as generative models have become increasingly capable. Early work by Wang et al. [23] demonstrated that a ResNet-based classifier trained on images from a single GAN generator could generalize surprisingly well to images produced by other unseen generators, establishing that CNN-generated images carry detectable statistical fingerprints. However, this finding was specific to GAN-based synthesis; the rapid rise of diffusion models and autoregressive generators has introduced new challenges that frequency-based or CNN-artifact-focused methods struggle to address.

More recent approaches have explored the use of large pretrained vision models. Ojha et al. [16] proposed using the feature space of CLIP as a universal representation for fake image detection, achieving strong cross-generator generalization without retraining on each new generator. The OPENFAKE benchmark [11], which serves as the primary training corpus in this work, was introduced specifically to address the gap between academic detection benchmarks and real-world politically motivated synthetic imagery, pairing real images with outputs from state-of-the-art generators including GPT Image 1, Imagen 3, Flux, and Stable Diffusion. Parallel large-scale benchmarks such as GenImage [27] have similarly pushed the field toward million-scale evaluation. In contrast to prior single-architecture evaluations, this work systematically compares three transformer-based detectors, ViT-B/16 in two training configurations and Swin Transformer V2-Base, on the same dataset, and integrates the best-performing model into a broader multimodal pipeline. The Vision Transformer (ViT) [5] established the foundation for attention-based image recognition by treating an image as a sequence of fixed-size patches, but its global self-attention incurs quadratic computational cost and lacks the inductive bias of locality. The Swin Transformer [28] addressed these limitations by restricting self-attention to local non-overlapping windows and introducing a shifted-window mechanism between consecutive layers to enable cross-window information exchange, yielding hierarchical multi-scale feature maps at linear complexity. Data-efficient training of vision transformers (DeiT) [29] further demonstrated that ViT-scale models can be trained competitively on ImageNet-scale data through knowledge distillation, without requiring the massive JFT-300M pretraining corpus used in the original ViT work.

B. Image Forgery Localization

Image-level classification determines whether an image is real or fake as a whole, but it provides no information about which specific regions have been altered. Pixel-level forgery localization is a significantly harder problem that requires the model to simultaneously reason about local inconsistencies and global semantic context. Classical benchmarks such as CASIA [4] and COVERAGE [24] established standard evaluation protocols for splicing and copy-move detection, while the Columbia dataset provided early baselines for uncompressed splicing. Real-world datasets such as NIST16 introduced more realistic manipulations collected from the internet.

On the modeling side, Huh et al. [8] proposed learning EXIF metadata consistency as a self-supervised proxy for detecting spliced regions, while more recent work has explored fusing RGB features with auxiliary modalities such as noise residuals and JPEG compression artifacts to improve localization accuracy [6], [10]. The DIS25k framework [21], which underlies the 100,000-image training corpus used in this work, introduced an automated pipeline for generating realistically spliced images at scale using object placement assessment, deep image matting via MatteFormer [17], and image harmonization via Harmonizer [9], addressing the scarcity of large annotated forgery datasets. This work builds on DIS25k by training AXUNet, which combines an Xception encoder with a Transformer self-attention bottleneck and a UNet++ nested decoder [26], a combination not previously explored in the forgery localization literature.

C. Automated Fact Verification

Automated fact verification has been formalized as a natural language inference task, where a model must determine whether a piece of textual evidence supports or refutes a given claim. The FEVER dataset [22] established the dominant benchmark for this task, pairing Wikipedia-sourced evidence with human-annotated claims and enabling direct evaluation of models' ability to reason over factual content. The SNLI corpus [1] and the broader MNLI benchmark [25] provided large-scale training data for general NLI, enabling cross-domain generalization studies.

The introduction of BERT [3] marked a turning point for NLI and fact verification, as bidirectional pretraining on large corpora allowed fine-tuned models to achieve near-human performance on several benchmarks. Subsequent improvements, including RoBERTa's removal of the next-sentence prediction objective and its training on substantially more data [13], and DeBERTa-v3's disentangled attention mechanism and replaced-token-detection pretraining [7], have progressively raised the state of the art. This work fine-tunes all three architectures under identical experimental conditions on a combined FEVER and SNLI corpus, providing a direct controlled comparison and identifying DeBERTa-v3-base as the strongest performer with 91.86% macro F1.

D. Explainability in Detection Systems

A persistent limitation of deep learning-based detection systems is their opacity: a model may correctly classify an image or claim as fake without providing any reasoning that a user can verify or trust. Explainability methods such as Grad-CAM [20] and LIME [18] have been widely applied to generate visual saliency maps that highlight image regions most influential to a classification decision. In the context of misinformation detection, however, such technical visualizations remain inaccessible to non-expert users. RealEyes addresses this gap by incorporating an explainability layer that translates model outputs, including confidence scores and pixel-level localization maps, into plain-language reasoning, a design choice not present in any of the prior single-modality systems discussed above.

III. Methodology

A. System Overview

RealEyes is a multimodal disinformation detection system composed of four specialized modules that operate in a coordinated pipeline. Given an input that may contain an image, a text claim, or both, the system routes each modality through its corresponding detection component and aggregates the outputs through a shared explainability layer that produces a unified, human-readable verdict. Figure 1 illustrates the overall architecture.

The image analysis branch consists of two complementary components. The first is an image-level binary classifier that determines whether a given image was produced by a generative AI model or captured by a real camera. The second is a pixel-level forgery localization model that identifies and spatially delineates regions within an image that have been composited or manipulated. These two components address different threat scenarios: the classifier handles fully synthetic images, while the localization model targets partially manipulated ones. The text analysis branch accepts a factual claim paired with supporting evidence and performs natural language inference to determine whether the evidence supports or refutes the claim. Finally, the explainability layer receives the outputs of all active modules and translates confidence scores, classification labels, and segmentation maps into plain-language reasoning accessible to non-technical users.

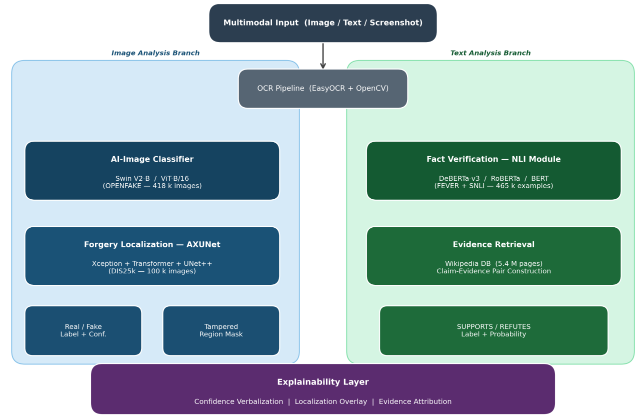

Fig. 1. Overview of the RealEyes multimodal detection pipeline. Inputs are routed through the image classification, forgery localization, and NLI modules, with outputs aggregated by the explainability layer.

B. AI-Generated Image Classification

The image classification module addresses the binary task of determining whether an input image is a real photograph or a fully AI-generated synthetic image. We formulate this as a supervised binary classification problem and train three transformer-based vision architectures, evaluating each under controlled conditions on the OPENFAKE benchmark.

The first two models are variants of the Vision Transformer (ViT-B/16), originally introduced by Dosovitskiy et al. [5]. ViT partitions an input image of resolution 224×224 into fixed-size non-overlapping patches of 16×16 pixels, linearly embeds each patch, and processes the resulting sequence with a stack of standard Transformer encoder blocks. A learnable `[CLS]` token is prepended to the sequence, and its final hidden state is passed to a classification head. The first variant (ViT v1) replaces the original ImageNet classification head with a single linear layer mapping to two output logits and fine-tunes all parameters end-to-end. The second variant (ViT v2) introduces additional regularization: the first six of twelve encoder blocks are frozen during training, a dropout layer with probability 0.3 is inserted before the classification head, label smoothing with $\varepsilon = 0.1$ is applied to the cross-entropy loss, and stronger data augmentation is used. This partial freezing strategy preserves low-level ImageNet representations while allowing high-level blocks to adapt to the real-versus-fake distinction.

The third model is Swin Transformer V2-Base (Swin V2-B), introduced by Liu et al. [14]. Unlike ViT's global self-attention, Swin computes attention within local shifted windows, producing hierarchical multi-scale feature maps and achieving linear computational complexity with respect to image size. Swin V2 additionally introduces scaled cosine attention and a log-spaced continuous position bias, improving stability when transferring from lower to higher resolutions. The model accepts 256×256 inputs, and its classification head is replaced with a dropout layer followed by a linear projection to two logits. Mixed-precision training is enabled via PyTorch's automatic mixed precision to accelerate training on GPU.

We additionally evaluate a higher-capacity variant of the Swin Transformer V2-Base(fourth model), pre-trained on ImageNet-22K and subsequently fine-tuned on ImageNet-1K, with shifted window sizes scaled progressively from 12 to 24 and input resolution scaled from 192 to 384 pixels. Compared to the standard Swin V2-B used in the three-way comparison, this variant introduces multi-scale pre-training, a larger pre-training corpus, and a higher-resolution fine-tuning stage, yielding 88.59 million parameters. An Exponential Moving Average (EMA) of model weights is maintained throughout training and used exclusively during validation, producing smoothed weight trajectories that improve generalization:

$$\theta_{\text{EMA}}(t) = \alpha \cdot \theta_{\text{EMA}}(t-1) + (1-\alpha) \cdot \theta(t) \tag{1}$$

where $\theta(t)$ are the current model weights and $\alpha \approx 0.999$ is the decay factor. The model is trained on a combined corpus of 762,769 images (381,694 real, 381,075 fake) with 686,493 training samples and 76,276 validation samples (approximately 90/10 split). All validation metrics for this variant are computed on the EMA model.

All four models are trained with the AdamW optimizer [15] and a cosine annealing learning rate schedule. The training objective is the cross-entropy loss, defined for a batch of N samples as:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=0}^{1} y_{i,c} \cdot \log\left(q_{i,c}\right) \tag{2}$$

where $y_{i,c} \in \{0,1\}$ is the one-hot ground-truth label for sample i and class c, and $q_{i,c}$ is the corresponding softmax probability produced by the model. For ViT v2 and Swin V2-B, label smoothing replaces the one-hot target with a smoothed distribution:

$$\tilde{y}_{i,c}^{\text{LS}} = (1-\varepsilon)\, y_{i,c} + \frac{\varepsilon}{C} \tag{3}$$

where $\varepsilon = 0.1$ and $C = 2$ is the number of classes. The cosine annealing schedule adjusts the learning rate at step t as:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left[1 + \cos\left(\frac{\pi t}{T_{\max}}\right)\right] \tag{4}$$

C. Video Classification

To extend the image classification module to video content, RealEyes incorporates a frame-level video analysis pipeline. Given an input video, the system samples frames at approximately one frame per second, clamped between a minimum of 8 and a maximum of 60 frames to handle both short clips and long videos. Each sampled frame is passed independently through the Swin V2-B (ms) image classifier, producing a real/fake probability pair per frame. The final verdict is determined by averaging the fake probabilities across all sampled frames:

$$\bar{p}_{\text{fake}} = \frac{1}{N} \sum_{i=1}^{N} p_{\text{fake}}^{(i)} \tag{5}$$

where N is the number of sampled frames and $p_{\text{fake}}^{(i)}$ is the predicted fake probability for frame i. The video is classified as AI-generated if $\bar{p}_{\text{fake}} \geq 0.5$, and as real otherwise. This approach allows the system to detect fully synthetic videos as well as partially manipulated ones, where only a subset of frames may exhibit generation artifacts.

D. Pixel-Level Forgery Localization

While the classification module determines whether an entire image is synthetic, many real-world manipulations involve only partial edits, a foreground object composited into a different background, for example. The localization module addresses this scenario by producing a pixel-wise probability map over the input image, identifying which specific regions have been manipulated.

We introduce AXUNet (Attention × Xception × UNet++), a U-shaped encoder-decoder architecture designed for binary segmentation of forged regions. Given an input RGB image $I \in \mathbb{R}^{H \times W \times 3}$, AXUNet produces a probability map $M \in [0,1]^{H \times W}$, where each value represents the likelihood that the corresponding pixel belongs to a manipulated region. A binary mask is obtained by thresholding M at $\tau = 0.5$.

The encoder is an ImageNet-pretrained Xception network [2], loaded via the `timm` library with `features_only=True`. Xception's depthwise-separable convolutions make it an effective texture discriminator, which is particularly valuable for forgery detection because manipulation artifacts frequently manifest as subtle local texture and frequency anomalies at region boundaries. The encoder produces multi-scale feature maps at five levels with channel dimensions of 64, 128, 256, 728, and 2048.

The deepest feature map is passed through a Transformer self-attention bottleneck before decoding. This bottleneck consists of two stacked Transformer encoder layers with 8 attention heads, GELU activation, a feed-forward dimension of $4C$ where C is the channel count, dropout of 0.1, and a learned positional embedding. The self-attention mechanism allows the network to compare distant spatial regions, capturing the global inconsistencies that characterize realistic splicing forgeries,lighting direction mismatches between a composited foreground and its background.

The decoder follows the UNet++ architecture [26], which extends the standard UNet skip connections with a nested, densely connected structure. Intermediate nodes $X^{i,j}$ in the decoder grid aggregate feature maps from all preceding nodes at the same resolution level, enabling multi-scale feature fusion and improving segmentation of small or irregularly shaped manipulated regions. The decoder progressively upsamples the encoded features through transposed convolutions, and a final 1×1 convolution projects the output to a single-channel logit that is bilinearly upsampled to the original input resolution.

The model is trained with a combined loss. The binary cross-entropy loss is:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log p_i + (1-y_i)\log(1-p_i)\right] \tag{6}$$

The Dice loss is:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon} \tag{7}$$

where $\epsilon = 1.0$ is a smoothing constant. The Focal loss variant is:

$$\mathcal{L}_{\text{Focal}} = -\alpha_t (1-p_t)^{\gamma} \log(p_t) \tag{8}$$

where $\alpha = 0.25$, $\gamma = 2.0$, and $p_t = p_i$ if $y_i = 1$, else $1 - p_i$. The combined losses used in training are:

$$\mathcal{L}_{\text{v1}} = 0.5 \cdot \mathcal{L}_{\text{BCE}} + 0.5 \cdot \mathcal{L}_{\text{Dice}} \tag{9}$$

$$\mathcal{L}_{\text{FD}} = 0.5 \cdot \mathcal{L}_{\text{Focal}} + 0.5 \cdot \mathcal{L}_{\text{Dice}} \tag{10}$$

E. Textual Claim Verification

The text verification module frames fact-checking as a 3-class natural language inference task. Given a piece of textual evidence e and a factual claim c, the module predicts whether e supports, refutes, or provides insufficient information to evaluate c. We fine-tune three pretrained transformer-based language models on this task: BERT-base-uncased [3], RoBERTa-base [13], and DeBERTa-v3-base [7]. All three models are loaded via the Hugging Face Transformers library using the AutoModelForSequenceClassification interface, which appends a linear classification head on top of the pooled [CLS] token representation, projecting from the hidden dimension of 768 to three output logits. The evidence and claim are concatenated as a sentence pair and tokenized with a maximum sequence length of 256 tokens. The full model, including all Transformer layers and the classification head, is fine-tuned end-to-end.

Training uses weighted cross-entropy loss with inverse-frequency class weights to compensate for label imbalance in the combined training corpus. For a single example with ground-truth class y and predicted probabilities p, the weighted loss is:

$$\mathcal{L}(y, p) = -\sum_{c} w_c \cdot y_c \cdot \log(p_c) \tag{11}$$

where $w_c = N_{\text{total}}/N_c$ is the weight for class c, with $w_{\text{SUPPORTS}} = 2.460$, $w_{\text{REFUTES}} = 3.105$, and $w_{\text{NEI}} = 3.682$. A linear learning rate schedule with warmup is applied:

$$\eta(t) = \eta_{\max} \cdot \min\left(\frac{t}{t_{\text{warm}}},\ \frac{T-t}{T-t_{\text{warm}}}\right) \tag{12}$$

where T is the total training steps, $t_{\text{warm}} = 0.10 \times T$, and $\eta_{\max} = 2 \times 10^{-5}$.

DeBERTa-v3 is distinguished from the other two models by its disentangled attention mechanism [7], which computes attention scores between tokens i and j as a sum of three components:

$$A_{ij} = A_{ij}^{c2c} + A_{ij}^{c2p} + A_{ij}^{p2c} \tag{13}$$

where A^{c2c} is the standard content-to-content attention score, A^{c2p} captures how the content of token i attends to the relative position of token j, and A^{p2c} captures how the position of token i attends to the content of token j.

F. Explainability Layer

A core design principle of RealEyes is that detection decisions must be interpretable by users without technical expertise. The explainability layer receives the outputs of all active modules, including classification labels, confidence scores, and pixel-level segmentation maps, and generates a plain-language summary of the evidence for each decision. For image inputs, the spatial localization map produced by AXUNet is overlaid on the original image to visually highlight tampered regions, while the classification confidence score is verbalized as a natural language statement. For text inputs, the NLI module's predicted label and probability distribution are translated into an explanation referencing the specific evidence sentence most relevant to the verdict. This design ensures that RealEyes is usable in practical contexts such as journalism, content moderation, and public fact-checking platforms where end users may not be familiar with machine learning terminology.

IV. Datasets

RealEyes is trained and evaluated using three distinct datasets, each corresponding to a specific detection module. Table I provides a summary overview. The following subsections describe each dataset in detail.

TABLE I
SUMMARY OF DATASETS USED ACROSS REALEYES MODULES

Dataset	**Task**	**Total**	**Train**	**Val/Test**
OPENFAKE	AI image classif.	418,524	334,820	83,704
DIS25k (ext.)	Forgery local.	100,000	95,000	5,000
FEVER + SNLI	Fact verif.	∼538,000	∼418,500	∼46,500

A. OPENFAKE: AI-Generated Image Classification

The image classification module is trained on OPENFAKE [11], a large-scale benchmark introduced specifically for real-versus-fake image detection in politically sensitive contexts. Real images are sourced from large-scale image-text corpora, including LAION, and from politically relevant social media scrapes. Synthetic counterparts are generated using state-of-the-art generative models, including GPT Image 1, Imagen 3, Flux, and Stable Diffusion versions 2.1 and 3.5. The dataset contains 418,524 images partitioned into 334,820 training images and 83,704 validation images following an 80/20 random split with seed 42. OPENFAKE was selected over alternative benchmarks such as GenImage [27] and FaceForensics++ [19] because it uniquely targets the current generation of diffusion and autoregressive image synthesizers rather than older GAN-based generators, and because its politically grounded real image collection more closely reflects the real-world distribution of images that a disinformation detection system would encounter in deployment..

B. DIS25k Extended: Pixel-Level Forgery Localization

The forgery localization module is trained on an extended version of the DIS25k framework [21]. The pipeline extracts foreground objects using MatteFormer [17], places them into background scenes using the Object Placement Assessment dataset [12], and applies Harmonizer [9] to reconcile lighting and color statistics. The extended corpus contains 100,000 images split into 95,000 training samples and 5,000 held-out test samples. All samples are resized to 512×768 pixels and normalized using ImageNet statistics.

C. FEVER and SNLI: Textual Claim Verification

The fact verification module is trained on a combination of FEVER [22] and SNLI [1]. FEVER contains human-annotated factual claims paired with Wikipedia evidence sentences. SNLI contributes 366,603 binary training examples after removing neutral pairs. The combined corpus contains approximately 538,000 training examples. For cross-domain evaluation, the MNLI corpus [25] is used solely as a held-out test set.

V. Implementation Details and Training

A. General Setup

All models were implemented in PyTorch and trained on CUDA-enabled GPUs. Vision models were built using the torchvision model zoo and the `timm` library. Language models were loaded via the Hugging Face Transformers library. A fixed random seed of 42 was used across all experiments to ensure reproducibility of data splits and stochastic training operations..

B. Image Classification Training

Table II summarizes the training configuration for all three image classifiers.

TABLE II
Training Configuration for Image Classification Models

Component	ViT v1	ViT v2	Swin V2-B	Swin V2-B (ms)
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning rate	1×10^{-4}	5×10^{-5}	1×10^{-4}	5×10^{-5}
Weight decay	1×10^{-4}	1×10^{-4}	1×10^{-4}	1×10^{-4}
Scheduler	Cosine	Cosine	Cosine	Cosine+WU
Loss	CE	CE+LS	CE+LS	BCE
Batch size	32	32	64	–
Max epochs	30	30	100	30
Early stopping	No	Pat.=5	No	No
Resolution	224^2	224^2	256^2	384^2
Mixed prec.	No	No	AMP	AMP
EMA	No	No	No	Yes
Dataset	OPENFAKE	OPENFAKE	OPENFAKE	Comb. 762k

The Swin V2-B (ms) variant differs from the standard Swin V2-B in three principal respects: it is initialized from a larger-scale multi-scale pretraining checkpoint (ImageNet-22K followed by ImageNet-1K fine-tuning at 384×384 resolution), it employs EMA weight averaging for all validation measurements as defined in Equation (4), and it is trained on an extended combined corpus of 762,769 images rather than the OPENFAKE benchmark used for the three-way comparison. A linear warm-up is applied during the first portion of training, with the learning rate increasing from approximately 1.17×10 to a peak of approximately 4.94×10 before following a cosine decay schedule.

The cross-entropy loss used as the base training objective is defined in Equation 1. For ViT v2 and Swin V2-B, the one-hot targets were replaced with smoothed distributions as defined in Equation 2, with a smoothing factor of 0.1. The learning rate followed a cosine annealing schedule as defined in Equation 3 for all three models. ViT v1 was fine-tuned end-to-end with all parameters trainable, serving as the full fine-tuning baseline. ViT v2 introduced four regularization mechanisms simultaneously: partial freezing of the first six encoder blocks, a dropout layer with probability 0.3 before the classification head, label smoothing, and stronger data augmentation including random rotation up to 10 degrees, stronger ColorJitter, and RandomErasing with probability 0.3. Early stopping was applied with a patience of five epochs monitored on validation F1, terminating training at epoch 10. Swin V2-B was trained with all parameters unfrozen and mixed-precision training enabled via PyTorch's automatic mixed precision (AMP) with a GradScaler, which roughly halved memory consumption and accelerated training without meaningful loss in accuracy. The Swin model accepted higher-resolution inputs of 256×256 pixels to match the resolution requirements of its pretrained weights.

C. Forgery Localization Training

Three variants of AXUNet were trained on the extended DIS25k corpus. All variants used the AdamW optimizer with a learning rate of 1×10 and a cosine annealing schedule. The Xception encoder was initialized from ImageNet pretrained weights loaded via timm, while the Transformer bottleneck and UNet++ decoder were trained from random initialization. All parameters were updated end-to-end throughout training without any frozen layers. Table 3 summarizes the training configuration across AXUNet variants.

Table III summarizes the training configuration across AX-UNet variants.

TABLE III
TRAINING CONFIGURATION FOR AXUNET VARIANTS

Component	v1 BCE	v1 Focal	v2
Optimizer	AdamW	AdamW	AdamW
Learning rate	1×10^{-4}	1×10^{-4}	1×10^{-4}
Loss	BCE+Dice	Focal+Dice	Focal+Dice
Resolution	512×768	512×768	512×768
Mixed prec.	No	No	AMP
Training	Single GPU	Single GPU	DDP
Augmentation	Basic	Basic	Albumentations

The combined loss function used in AXUNet v1 sums binary cross-entropy and Dice loss to jointly optimize pixel-wise classification accuracy and region-level overlap. Dice loss addresses the severe class imbalance inherent in forgery segmentation, where manipulated pixels typically constitute a small fraction of the total image area. The Focal loss variant, used in the second configuration and in v2, further down-weights the contribution of easily classified background pixels during training, directing gradient updates toward the harder foreground forgery regions. AXUNet v2 additionally employed a distributed data parallel training strategy and an Albumentations-based augmentation pipeline including random flips, rotations, brightness and contrast adjustments, and grid distortions to improve generalization to unseen manipulation patterns.

D. Fact Verification Training

All three language models were fine-tuned under identical hyperparameter settings to ensure a controlled and fair comparison. Table 4 summarizes the shared training configuration.

Table IV summarizes the shared training configuration for all three language models.

TABLE IV
SHARED TRAINING CONFIGURATION FOR FACT VERIFICATION MODELS

Hyperparameter	Value
Optimizer	AdamW
Learning rate	2×10^{-5}
Warmup ratio	10% of total steps
Batch size	64
Maximum epochs	6
Early stopping pat.	3 epochs (val macro F1)
Max sequence length	256 tokens
Gradient clipping	Max norm = 1.0
Label smoothing	$\varepsilon = 0.05$
Mixed precision	FP16 via PyTorch AMP
Random seed	42

Training used the weighted cross-entropy loss defined in Equation 4, with class weights computed over the combined FEVER and SNLI training corpus as described in Section 4.3. The linear learning rate schedule with warmup defined in Equation 5 was applied across all three models to stabilize early training, when the randomly initialized classification head produces large and noisy gradients. Gradient clipping with a maximum norm of 1.0 was applied before each optimizer step to prevent gradient explosion in the early warmup phase. Early stopping monitored validation macro F1 with a patience of three epochs, saving the best model checkpoint to disk at each improvement. After training concluded, the best saved checkpoint was loaded for all evaluation on the FEVER blind test set and the MNLI cross-domain evaluation splits. The three models differ only in their pretrained backbone and tokenizer. BERT-base-uncased uses a WordPiece tokenizer and bidirectional masked language model pretraining. RoBERTa-base uses a byte-pair encoding tokenizer and was pretrained with dynamic masking on a substantially larger corpus. DeBERTa-v3-base uses a SentencePiece tokenizer and incorporates the disentangled attention mechanism defined in Equation 6, which computes separate content-to-content, content-to-position, and position-to-content attention scores between every token pair.

VI. RESULTS

A. AI-Generated Image Classification

Table V presents the best validation performance of each image classification model on the OPENFAKE benchmark.

TABLE V
BEST VALIDATION PERFORMANCE ON OPENFAKE

Model	Acc.	Recall	F1	Ep.
ViT-B/16 v1	94.05%	0.9407	0.9404	30
ViT-B/16 v2	94.56%	0.9533	0.9459	9
Swin V2-B	**98.18%**	**0.9826**	**0.9818**	**12**

Swin V2-B substantially outperformed both ViT variants, achieving a validation accuracy of 98.18% and an F1 score of 0.9818 in only 12 epochs, roughly one-third of the training budget required by ViT v1. This performance gap of approximately 4 percentage points in both accuracy and F1 is consistent with Swin's architectural advantages: its hierarchical multi-scale feature maps and shifted-window attention mechanism are better suited to the local texture artifacts that distinguish AI-generated images from real photographs than ViT's global patch-level attention.

Comparing the two ViT configurations, ViT v2 with partial freezing, dropout regularization, and label smoothing achieved higher validation accuracy (94.56% vs. 94.05%) and recall (0.9533 vs. 0.9407) than full fine-tuning despite training for only 9 epochs before early stopping. ViT v1 exhibited a growing generalization gap across its 30 training epochs, with training accuracy reaching 99.99% while validation accuracy plateaued at 94.05%, a classic overfitting signature consistent with full fine-tuning on a large dataset without sufficient regularization. Swin V2-B demonstrated the most stable training dynamics among the three models, with training and validation accuracy converging closely throughout the logged 14 epochs and no sign of divergence.

TABLE VI
PER-EPOCH PERFORMANCE OF SWIN V2-B (MS) ON COMBINED CORPUS

Epoch	Split	Loss	Acc.	Prec.	Rec.	F1
1	Train	0.3744	86.18%	86.20%	86.15%	86.17%
1	Val (EMA)	0.0883	98.56%	99.34%	97.77%	98.55%
2	Train	0.3402	88.83%	88.79%	88.89%	88.84%
2	Val (EMA)	0.0814	98.54%	99.44%	97.61%	98.52%
3	Train	0.3381	89.07%	89.00%	89.15%	89.08%
3	Val (EMA)	0.0817	98.67%	99.38%	97.94%	98.66%
4	Train	0.3314	89.20%	89.19%	89.21%	89.20%
4	**Val (EMA)** ⋆	**0.0748**	**99.02%**	**99.45%**	**98.58%**	**99.01%**

Best checkpoint saved (Val F1 = 0.9901). All validation metrics computed on EMA model. The Swin V2-B (ms) variant achieves a best validation F1 of 0.9901 and accuracy of 99.02% by Epoch 4, surpassing both the standard Swin V2-B (F1 = 0.9818) and all ViT configurations. The training loss decreases monotonically from 0.3744 to 0.3314, while validation F1 improves consistently, with no sign of overfitting over the observed window. Validation performance consistently exceeds training performance across all epochs — a pattern attributable to the use of EMA weights during evaluation combined with data augmentation applied only during training. Precision exceeds recall in all validation epochs (e.g., 99.45% vs. 98.58% at Epoch 4), indicating a mild conservative bias toward avoiding false positive forgery detections, which is desirable in forensic contexts.

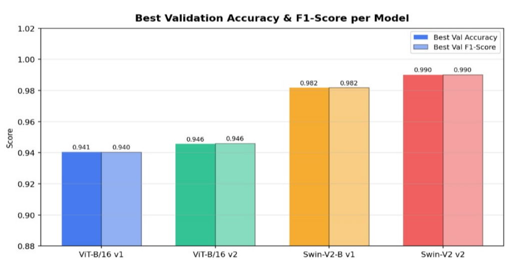

Fig. 2. Best Validation Accuracy F1-Score per Model

B. Pixel-Level Forgery Localization

Table VII reports the best validation results for each AXUNet variant on the 5,000-image held-out test split.

TABLE VII
BEST VALIDATION PERFORMANCE OF AXUNET VARIANTS ON DIS25K

Variant	Ep.	IoU	F1	Prec.	Rec.
v1 BCE+Dice	46	0.8999	0.9461	n/a	n/a
Focal+Dice	**21**	**0.9034**	**0.9485**	**0.9619**	**0.9370**
v2 Aug+DDP	48	0.8936	0.9401	n/a	n/a

The Focal+Dice variant achieved the strongest validation performance across all metrics, reaching an IoU of 0.9034 and F1 of 0.9485 by epoch 21, converging more than twice as fast as the BCE+Dice baseline, which required 46 epochs to reach its best IoU of 0.8999. This faster and superior convergence is consistent with the design principle of focal loss: by down-weighting the contribution of easily classified background pixels, gradient updates are concentrated on the small, hard-to-segment manipulated foreground regions that represent the core challenge in forgery localization. The precision of 0.9619 paired with a recall of 0.9370 in the Focal+Dice variant indicates a mild conservative bias, the model is slightly more likely to miss a small number of manipulated pixels than to incorrectly flag authentic regions as forged. In forensic applications such as media verification and legal evidence analysis, this conservative behavior is preferable: a false accusation of forgery on an authentic image carries greater real-world cost than a missed detection. AXUNet v2, despite its richer augmentation pipeline, mixed-precision training, and distributed data parallel setup, produced slightly lower validation IoU (0.8936) in the 49 logged epochs compared to both v1 configurations. This result suggests that the stronger geometric and photometric augmentations applied in v2 may have disrupted the subtle low-level frequency and texture cues that the Xception encoder relies upon for forgery detection.

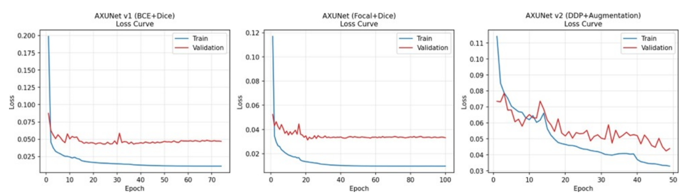

Fig. 3. Training and validation loss curves for all three AXUNet variants.

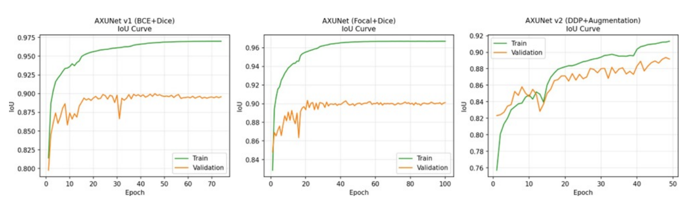

Fig. 4. Per-epoch validation IoU curves for all three AXUNet variants.

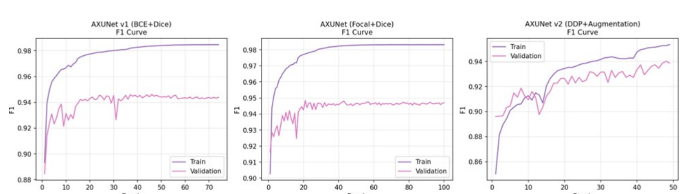

Fig. 5. Per-epoch validation F1 curves for all three AXUNet variants.

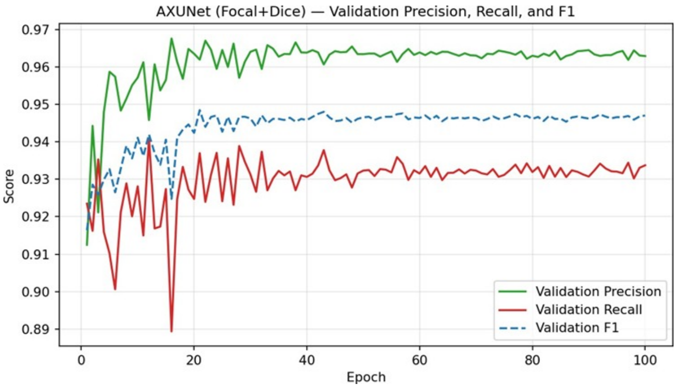

Fig. 6. Validation precision, recall, and F1 across all 100 epochs of the Focal+Dice variant.

Figure 6 reveals that the Focal+Dice model's precision stabilizes at approximately 0.96 by convergence, while recall stabilizes at approximately 0.93, confirming a mild conservative detection bias. The precision and recall curves diverge early in training and maintain their relative ordering throughout all 100 epochs, suggesting this is a structural property of the model rather than a training artifact.

C. Textual Claim Verification

1) Internal Validation Performance: Table VIII reports the best validation performance of each language model on the internal FEVER validation split.

TABLE VIII
BEST VALIDATION PERFORMANCE ON FEVER INTERNAL SPLIT

Model	Acc.	Prec.	Rec.	F1
BERT-base	88.81%	88.39%	88.50%	88.44%
RoBERTa-base	90.74%	90.36%	90.48%	90.41%
DeBERTa-v3	**92.14%**	**91.85%**	**91.87%**	**91.86%**

TABLE IX
TRAINING LOSS AND VALIDATION MACRO F1 PER EPOCH

Ep.	B-Loss	B-F1	R-Loss	R-F1	D-Loss	D-F1
1	0.6574	0.8476	0.5725	0.8897	0.5506	0.9057
2	0.4795	0.8701	0.4167	0.8977	0.3727	0.9126
3	0.4358	0.8776	0.3857	0.9013	0.3527	0.9162
4	0.4130	0.8818	0.3676	0.9023	0.3395	0.9172
5	0.3988	0.8835	0.3585	0.9030	0.3301	0.9180
6	0.3909	0.8844	0.3505	0.9041	0.3241	0.9186

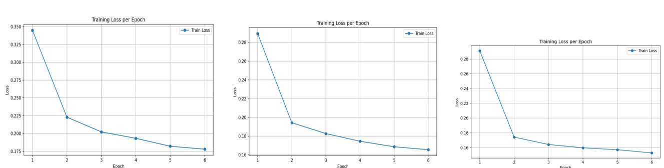

Fig. 7. Training loss per epoch for BERT (left), RoBERTa (center), and DeBERTa (right).

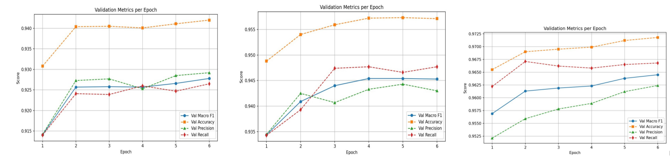

Fig. 8. Validation metrics per epoch for BERT (left), RoBERTa (center), and DeBERTa (right).

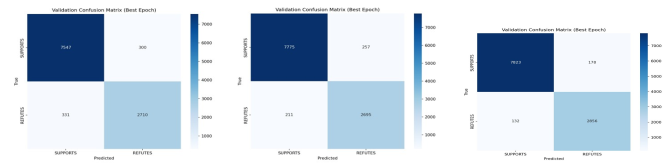

Fig. 9. Validation confusion matrices at best epoch for BERT (left), RoBERTa (center), and DeBERTa (right).

DeBERTa achieved a validation macro F1 of 0.9057 in its very first epoch, already outperforming BERT's best result across all six epochs, and continued improving to 0.9186 by epoch 6.

2) Blind Test Performance: Table X reports performance on the official FEVER shared-task development set of 13,332 examples.

TABLE X
BLIND TEST PERFORMANCE ON FEVER SHARED-TASK DEV SET

Model	Acc.	Prec.	Rec.	F1
BERT-base	88.51%	88.05%	88.17%	88.10%
RoBERTa-base	90.74%	90.30%	90.44%	90.36%
DeBERTa-v3	**92.09%**	**91.69%**	**91.76%**	**91.72%**

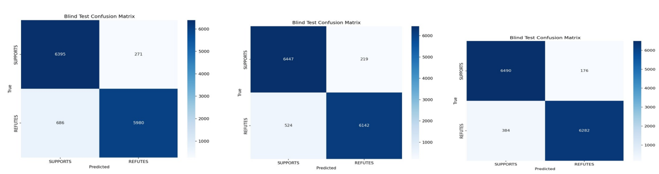

Fig. 10. Blind test confusion matrices for BERT (left), RoBERTa (center), and DeBERTa (right).

3) Cross-Domain Generalization on MNLI: Table XI reports performance on the MNLI matched and mismatched validation sets.

TABLE XI
CROSS-DOMAIN GENERALIZATION ON MNLI

Model	Matched F1	Mismatched F1	Gap
BERT-base	63.21%	64.02%	−25.2 pp
RoBERTa-base	77.29%	76.54%	−13.1 pp
DeBERTa-v3	**80.33%**	**79.45%**	**−11.4 pp**

The cross-domain results reveal a substantial performance gap across all three models when evaluated on MNLI, which is expected given the more challenging 3-class formulation applied in this evaluation. BERT suffers the largest drop of over 25 percentage points, indicating that its representations are strongly anchored to the Wikipedia and crowdsourced-caption style of text present in FEVER and SNLI. RoBERTa generalizes better but still shows a gap of 13.1 percentage points, confirming that even its more extensive pretraining corpus is not sufficient to fully bridge the 3-class domain shift. DeBERTa shows the smallest cross-domain gap at 11.4 percentage points, establishing that its disentangled attention mechanism captures more transferable syntactic and semantic structure than the other two models. Despite all three models showing lower MNLI scores than their FEVER blind test results, DeBERTa remains the strongest generalizer among the three, making it the most suitable choice for real-world deployment in RealEyes where claim verification spans many domains beyond Wikipedia.

D. Overall System Summary

Table XII provides a consolidated summary of the best-performing configuration in each RealEyes module.

TABLE XII
SUMMARY OF BEST-PERFORMING MODEL PER REALEYES MODULE

Module	**Best Model**	**Metric**	**Value**
Image Classification	Swin V2-B	Val F1	0.9818
Image Classification (ext.)	Swin V2-B (ms)	Val F1	0.9901
Forgery Localization	AXUNet Focal+Dice	Val IoU	0.9034
Fact Verif. (val)	DeBERTa-v3	Macro F1	91.86%
Fact Verif. (blind)	DeBERTa-v3	Macro F1	91.72%
Fact Verif. (MNLI)	DeBERTa-v3	Matched F1	80.33%

VII. CONCLUSION

This paper presented RealEyes, a multimodal disinformation detection system that unifies AI-generated image classification, pixel-level forgery localization, and automated fact verification within a single integrated pipeline augmented by a human-readable explainability layer. The system was motivated by the observation that modern disinformation rarely takes a purely textual or purely visual form, and that existing detection approaches address each modality in isolation without providing interpretable reasoning to end users.

Across all three detection modules, RealEyes achieved strong experimental results. The Swin Transformer V2-Base image classifier reached 98.18% validation accuracy and an F1 score of 0.9818 on the large-scale OPENFAKE benchmark, outperforming both ViT-B/16 configurations by a margin of approximately 4 percentage points while converging in only 12 epochs. The AXUNet forgery localization model, trained with a combined Focal and Dice loss, achieved a pixel-level IoU of 0.9034 and an F1 score of 0.9485 on the DIS25k test split, with precision and recall of 0.9619 and 0.9370 respectively. The DeBERTa-v3-base fact verification model achieved a macro F1 of 91.86% on the internal FEVER validation split and 91.72% on the held-out blind test set, with cross-domain generalization on MNLI reaching a matched F1 of 80.33%.

The current work focuses on module-level evaluation, with each component validated independently on its respective benchmark. As with any specialized system, performance on manipulation types outside the training distribution, such as inpainting or face-swap forgeries, and on generators released after the OPENFAKE dataset was constructed, represents a natural direction for future extension. Future work will explore joint fusion of image-level, pixel-level, and textual evidence into a unified inference architecture, expansion of the forgery localization module to a broader range of manipulation types, out-of-distribution robustness evaluation for the image classifier, and a user study assessing whether the explainability layer's outputs are genuinely trusted by non-technical users in realistic fact-checking scenarios.

Disinformation is one of the defining challenges of the current information environment, and the pace at which generative models are improving means that the gap between human perception and synthetic media will only widen. RealEyes demonstrates that multimodal detection, combining vision and language with pixel-level localization and explainable outputs, is a viable and principled approach to this problem. We hope this work contributes a useful foundation for future research and deployment of trustworthy disinformation detection systems.

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